

EPS Loyalty

Host Implementation Guide

December 15, 2017



Verifone®

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Verifone, Inc.

Verifone®, Inc.

88 West Plumeria Drive

San Jose, CA 95134

Telephone: 408-232-7800

<http://www.verifone.com>

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Revision History

Rev	Date	Description
1.00	6/20/13	Initial release.
1.01	7/31/13	Added definitions, more XML examples, receipt examples, Cash Sale with Loyalty flow diagram, and information setting discountable flags.
1.02	12/15/17	Added section 3.9 Customer Messages.

Audience

VeriFone Engineering	The primary audience for this document is VeriFone engineers responsible for implementing loyalty functionality in VIPER.
EPS Loyalty Host Partners	A primary audience for this document is EPS Loyalty Host partners as they move their support for VeriFone EPS and POS solutions to the PCATS standard.

Reference Documents

The following documents were referenced in the creation and maintenance of this document.

Document	Version
PCATS POS Loyalty Message Specification	1.0.1
Loyalty Schema	1.04
NAXML Core Schema	0.9

Definition of Terms

Term	Definition
DCR	Dispenser Card Reader
EPS	Electronic Payment System
FEP	Front End Processor
LHFEP	Loyalty Host Front End Processor
PCATS	Petroleum Convenience Alliance for Technology Standards

PCI-DSS	Payment Card Industry Data Security Standard
POS	Point of Sale
PPG	Price Per Gallon
PPU	Price Per Unit

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1. Introduction

This document specifies the functions of the EPS Loyalty Front End Processor (FEP). These functionalities include using the PCATS POS-Loyalty Host Specification, price-per-unit discounting, transactions-level discounting, and item level discounting.

The EPS can support communications to multiple loyalty partners in the same transaction. Whether this occurs depends on the POS and the level of software on the POS. Any POS local discounting is out of scope for the purposes of this document.

1.1 Multiple PCATS Loyalty Front End Processors

EPS can run multiple PCATS loyalty front end processors (FEPs). A combination of loyalty host connections can be established in the EPS, and is driven by FEP Tables. It is possible to have four combinations of loyalty FEP connections.

Each of the loyalty FEP modules will be running independently. Because each front-end-processor is running independently, it is also possible to have one host to be offline and the other online.

Enabling and disabling of the loyalty FEPs is completely controlled by the FEP Table.

1.2 Loyalty Host Front End Processor (LHFEP)

A new entry in the FEP Table is required to enable any LHFEP. Fields in the FEP Table will be populated differently between loyalty hosts. The parameters shown in the Test LH FEP Table entry example below are part of configuring and setting up an LFEP.

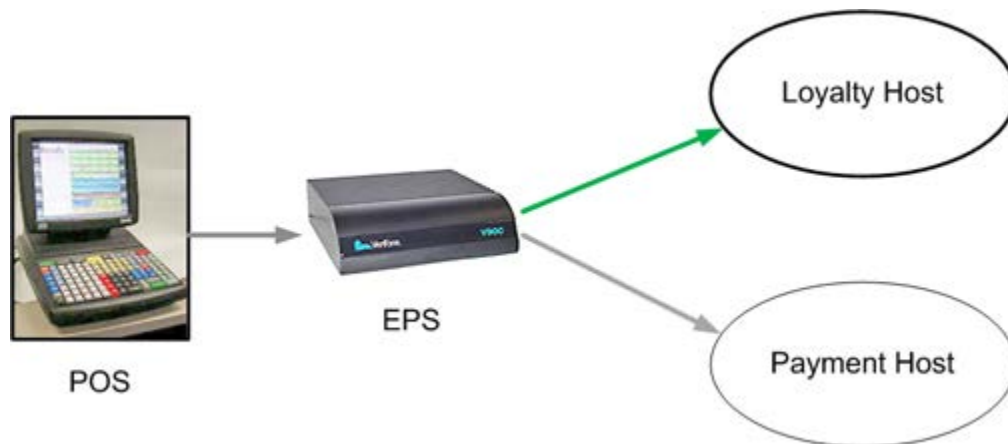
```
<FEPRow>
  <FEPName>TEST</FEPName>
    <IPAddress>192.168.31.200</IPAddress>
    <IPAddressPort>9010</IPAddressPort>
    <MerchantID>000125059999566</MerchantID>
    <TerminalID>01</TerminalID>
    <TimeSyncFlag>0</TimeSyncFlag>
    <AdminMessageFlag>0</AdminMessageFlag>
    <StoreAndForwardMaximum>50</StoreAndForwardMaximum>
    <MessageTimeout>15</MessageTimeout>
    <BatchSize>0</BatchSize>
    <ProductCodeHandlingRoutine>0</ProductCodeHandlingRoutine>
    <ProductBucketsSupported>0</ProductBucketsSupported>
    <RollupRulesRoutine>0</RollupRulesRoutine>
    <ActionCodeTableName>PCATSAActionCodeTable</ActionCodeTableName>
    <HeartbeatTimer>0</HeartbeatTimer>
    <OnlineToUpdate>>false</OnlineToUpdate>
    <CardMatchPrecedence>-1</CardMatchPrecedence>
    <ProgramName>TEST PROGRAM</ProgramName>
    <ProgramPrecedence>1</ProgramPrecedence>
    <LoyaltyPrompting>>false</LoyaltyPrompting>
    <FEPEnabled>>true</FEPEnabled>
</FEPRow>
```

Note: The FEPTable above is an example. The given attribute values are not the final values and they may change.

2. EPS Flow

This section discusses several aspects of EPS processing and logic.

The following diagram shows the processing flow for loyalty transactions. The scope of this document primarily describes the processing flow between the EPS and the Loyalty host



Loyalty ID Entry Methods

The EPS can acquire customer card information from either the POP (via a prompt) or the DCR.

- **Swipe Entry** – A customer can swipe their loyalty cards in POP/ DCR at any point in a transaction prior to payment (Cash/Credit).
- **Manual Entry** – Manual entry of loyalty is supported. This includes prompting for manual entry after three bad card swipes, as well as support for a manually entered alternate loyalty identifier.
- **Scan Entry** – Loyalty bar code scan can be supported, but this functionality still is not in scope.

Loyalty Track to Host

Track preference in EPS is controlled by Card Matching logic, which provides the clarity on which tracks are to be sent to host for a matched card. If a primary track is configured, only one track will be sent to the host.

The PCATS loyalty specification does not include a field to indicate which track was matched on a loyalty card swiped. This can cause an issue when a track 1 card is read and the card reader sends “junk” in the track 2 fields. To solve this, the EPS will always send only the matched track to the loyalty host.

3 Loyalty FEP Module Functions

This section details functionalities of the EPS Loyalty FEP Module. Multiple PCATS loyalty FEP modules can be run together. Based on the loyalty card swiped during a loyalty transaction, the corresponding loyalty FEP will be invoked to process the loyalty request.

3.1 EPS Loyalty FEP Module

The EPS Loyalty FEP module is the interface between EPS and the Loyalty FEP.

3.2 EPS Loyalty FEP Module and Loyalty Host Communication

The request /response messages sent and received from Loyalty Host will use XML over standard TCP/IP socket communication.

- EPS Loyalty FEP module will ensure the 'persistent' communication to the Loyalty host over standard non-encrypted TCP/IP socket connection.
- The connection between the EPS and the Loyalty Host supports a "heartbeat." The heartbeat signal uses 'GetLoyaltyOnlineStatus' message to identify whether the Loyalty Host is online or not. This heartbeat is an inactivity timer, which sends the 'GetLoyaltyOnlineStatusRequest' message to the Loyalty Host if no messages were sent from EPS to the Loyalty Host for a particular time period. This time period will be set based on the value obtained from the FEP Table.

If no response is received from the Loyalty host for any loyalty request message for the configured timeout period, then the Loyalty Host will be considered to be offline. The channel will go into a recovery mode. The following steps describe the actions taken when the channel is in recovery mode.

- a) Check if the connection to the loyalty host is still alive
- b) If connection is active and it is just a timeout from the host system, then proceed to next step. If connection lost, then try to reconnect to the loyalty host until it get connected to host system. After the connection re-establishes, go to the next step
- c) Send a heartbeat message 'GetLoyaltyOnlineStatusRequest' to the loyalty host to identify whether the host is online.
- d) Continue the above step until there is a response from the host. The timeout period used for 'GetLoyaltyOnlineStatusRequest' message comes from its FEPTable entry; the default timeout period is 15 seconds. Each time, check whether the connection is active. If the connection is lost in between, then try to reconnect until connection to the host system re-establishes. If you get a 'success' response for 'GetLoyaltyOnlineStatusRequest' message, then it means the connection is online and the channel will reset back to online mode.

In case of a response with 'OverallResult = Failure', the 'GetLoyaltyOnlineStatusRequest' message will be resent since it indicates that the message format was not proper. If a response has been received with 'OverallResult=Success', then the Loyalty Host will be determined to be online. But, whether the loyalty processing is available or not will be determined based on the 'PromptForLoyaltyFlag' in the 'GetLoyaltyOnlineResponse' message.

3.3 EPS Loyalty FEP Module and Loyalty Host Interface

EPS Loyalty FEP module will receive a request from the core EPS and from the requests (PCATS) to the Loyalty Host (FEP).

EPS Loyalty FEP module will implement the following request/response messages.

Refer to the *PCATS POS Loyalty Message Specification* for further details.

GetRewards

The GetRewards message uses the following request/response format.

GetRewardsRequest

Message Format

This section explains some of the elements of the GetRewardsRequest message that require additional/specific information. All the other elements which are not mentioned in this section can be referred from the PCATS POS Loyalty Specification document.

Field Name	Element /Attribute	Field Data
LoyaltyID	E	Unique Loyalty Identifier
FuelPriceData	E	This will be set only once during the lifetime of transaction [First Time only]. This will have the fuel price and grade id.
TransactionData	E	This will have the purchased product information. If the FuelPriceData is present, this element will not appear. [For example, Customer fuels outside (1 st GetRewardsRequest with FuelPriceData only) and purchases merchandise [pay inside](2 nd GetRewardsRequest with TransactionData only).
TenderInfo	E	This is the child element of 'TransactionLine' which is the child element of 'TransactionData'. This element contains transaction tender details. 'TenderInfo' will be set in 'GetRewards' message only for transaction level discounts.
TenderCode	E	This is the PCATS identifier for the tender value. This value will be 'loyaltyOffer', if the tender represents a transaction level loyalty discount.
ISOPrefix	E	The first 6 digits of an account number for card based method of payment.

GetRewardsResponse

Message Format

This section explains some of the elements of GetRewardsResponse message that require additional/specific information. All other elements not mentioned in this section can be obtained from the *PCATS POS Loyalty Specification* document.

Field Name	Element /Attribute	Field Data
ResponseHeader	E	It has an attribute 'overallResult', which can take the following values: success, failure, partialFailure, busy, timeout, parsingError, validationError, missingMandatoryData. Note: This element is common to all types of response messages.
Result		This element can take the following optional child elements:
Success		'overallResult' attribute should be 'success'.
Failure		'overallResult' attribute can take 'success', 'failure', 'busy', 'timedOut', 'parsingError', 'validationError' and 'missingMandatoryData'. The following scenarios can give the 'Result' element 'failure' even though the 'overallResult' is 'success'. In a transaction with loyalty, if the loyalty account is valid but contains no discount, in such case 'overallResult' value will be 'success'. In the event a message appears out of sequence. (for example, the first message received for a transaction is a 'CancelTransactionRequest'), the 'overallResult' value will be 'success'.
LoyaltyIDValidFlag	E	This will indicate the provided LoyaltyID is valid or not.

RewardActions	E	This element will have the PrepayFuelRewards or AddReward elements as child. RemoveReward element will not be supported
PrepayFuelRewards	E	This will have the discounts that can be applied to the fuel to be dispensed only. This is used for PPG discounts. This will have FuelGradeReward elements for each FuelPrice elements sent in the GetRewardsRequest.
RewardDiscountMethod	E	This is the child element of PrepayFuelRewards and AddReward elements. The only supported discount method is 'amountOff'.
RewardLimit	E	The maximum quantity /volume to which the reward can apply. <i>[If the quantity purchased is more than the reward limit, the discount will be applied up to the reward limit. RewardLimit =0 implies no quantity/volume limit for the discount]</i>
RewardsPromptTextShort/Long	E	Child Element to PrepayFuelReward and AddReward. This text is used to prompt the customer whether to apply the loyalty discount. Only the "RewardPromptTextShort" message will be sent back to POS .
AddReward	E	AddReward element contains information about item-level discounts and transaction-level discounts. If the fuel is already purchased, then PrepayFuelReward element would not be present. In such scenario, only AddReward element will be present.

Note: Customer swipes a local loyalty card (bin range overlaps with the national loyalty card), Loyalty Host will set the "LoyaltyIDValidFlag" attribute value ="no". EPS should forward the response along with the track details to the POS so that POS can validate the card for local loyalty. If the POS found a local loyalty card match, the POS shall give a discount on the transaction. The Loyalty Host would not process the local loyalty discount while finalizing the transaction

FinalizeRewards

FinalizeRewardsRequest

Message Format

This section explains some of the elements of the FinalizeRewardsRequest message that require additional/specific information. All other elements not mentioned in this section can be referenced in the *PCATS POS Loyalty Specification* document.

Field Name	Element /Attribute	Field Data
LoyaltyID	E	Unique Loyalty Identifier. If this tag does not exist, then a valid Loyalty ID was not presented during the transaction.
TransactionData	E	This will have the POS transaction details.
TenderInfo	E	This is the child element of 'TransactionData'. This element contains tender data.
TenderCode	E	This is the PCATS identifier for the tender value. This value will be 'loyaltyOffer' if the tender represents a transaction level loyalty discount.
TenderSubCode	E	CardCircuit
ISOPrefix	E	The first 6 digits of an account number for card-based method of payment.

FinalizeRewardsResponse

Message Format

This section explains some of the elements of the FinalizeRewardsResponse message that require additional/specific information. All the other elements not mentioned in this section can be reference in the *PCATS POS Loyalty Specification* document.

Field Name	Element /Attribute	Field Data
CutomerMessageData	E	This data is used to send receipt messages to POS.
DisplayData	E	Contains display command for the POS
DisplayCommand	E	It is the child element of 'DisplayData'. This will direct the POS to display text to the customer or cashier. It has the attributes informing about the target device, the time when the text should be displayed and the duration of the message.
DisplayLine	E	A single line of display text.

ReceiptData	E	This has the receipt lines for the POS to print in the receipt above the footer.
ReceiptLine	E	This is the child element of ReceiptData. This has a single line of receipt text. Note: This will not support the 'barCodeFormat' attribute specified in the PCATS message format.

CancelTransaction

CancelTransactionRequest

This request is created by EPS Loyalty FEP Module when the customer or cashier cancels/voids the transaction that is in progress before it is finalized.

Field Name	Element /Attribute	Field Data
RequestHeader	E	Used to inform the Loyalty System that the Customer or the Cashier has canceled or voided the transaction. The Loyalty System should cancel or roll back any Customer accruals or redemptions pending for the transaction at this point. This is normally an informational message, but a response is required to ensure that a stored and forwarded request has been successfully received.

CancelTransactionResponse

This response is required to ensure that a stored and forwarded request has been successfully received.

Field Name	Element /Attribute	Field Data
ResponseHeader	E	Contains the transaction-related data to ensure the store and forward request has been received for that transaction.

ReverseTransaction

ReverseTransactionRequest

This request is created by the EPS Loyalty FEP Module when the cashier cancels/voids the previously finalized [after FinalizeRewards] transaction. This message can also be called for the Loyalty refund transaction.

Field Name	Element /Attribute	Field Data
LinkedTransactionInfo	E	Used for a reversal of a previous transaction, this defines the transaction being reversed.
TransactionData	E	This will have the POS transaction details.

ReverseTransactionResponse

This response will ensure that the transaction has been reversed. It also contains the customer message data be provided to the customer.

Field Name	Element /Attribute	Field Data
CustomerMessageData	E	Display and/or Receipt messages to be provided to the Customer.

Note: Partial reversals are allowed by the loyalty host.

GetRewardStatus

This message is used for balance inquiries.

GetRewardStatusRequest

Message Format

This section explains some of the elements of the GetRewardStatusRequest message that requires additional/specific information. All other elements not mentioned in this section can be referenced in the *PCATS POS Loyalty Specification* document.

Field Name	Element /Attribute	Field Data
GetRewardStatusRequest	E	Has 'rewardStatusType' attribute that will support only the 'detail' as value.

GetRewardStatusResponse

Message Format

This section explains some of the elements of the GetRewardStatusResponse message that require additional/specific information. All other elements not mentioned in this section can be referenced in the *PCATS POS Loyalty Specification* document.

Filed Name	Element /Attribute	Field Data
CustomerMessageData	E	Data to display or print customer loyalty account details.

EndPeriod

EndPeriodRequest

Used to inform the Loyalty System that the POS system is ending a business period and beginning a new one. The 'PeriodTotals' element will not be present in the request as per the requirement.

Field Name	Element /Attribute	Field Data
BusinessPeriod	E	The business period being closed on the POS System.
PeriodTotals		POS System totals for the period.
LoyaltyReportMaxWidth		The maximum width in columns supported by the POS System for any loyalty period reports. This can be used, for example, to notify the Loyalty System that a 40-column printer will be used to output any loyalty period reports.

EndPeriodResponse

"LoyaltyReportLocation" element will not be used in the response message

GetLoyaltyOnlineStatus

This message is used to find whether the loyalty host is online or available. And also used for implementing a 'heartbeat' signal.

GetLoyaltyOnlineStatusResponse

Message Format

This section explains some of the elements of the GetRewardsRequest message that require additional/specific information. All other elements not mentioned in this section can be referenced in the *PCATS POS Loyalty Specification* document.

Filed Name	Element /Attribute	Field Data
PromptForloyaltyFlag	E	If this element has the value "yes" for the attribute 'value', it indicates that the Loyalty Host and all loyalty functions are available in the Loyalty Host. If the 'value' is "no", it indicates that the Loyalty Host is available but the loyalty functions are not available for processing.

CancelRedemptionMessage

This message is sent to loyalty host when a customer disapproves an optional reward offered by that host. It is just to inform the loyalty host that the optional reward has not been applied.

CancelRedemptionMessage

Message Format

This section explains some of the elements of the CancelRedemption message.

Filed Name	Element /Attribute	Field Data
LoyaltyRewardID	E	This is a unique identifier for each reward that is generated by a loyalty system.

Out of Scope Messages

The following messages are currently not supported by our LHFEP modules:

PCATS Message Title	Document References and Notes
BeginCustomer / EndCustomer	This message is considered as out of scope since the communication starts with GetRewards.
GetCustomerMessaging	This message is considered as out of scope.

3.4 Difference between CancelTransaction and ReversalTransaction

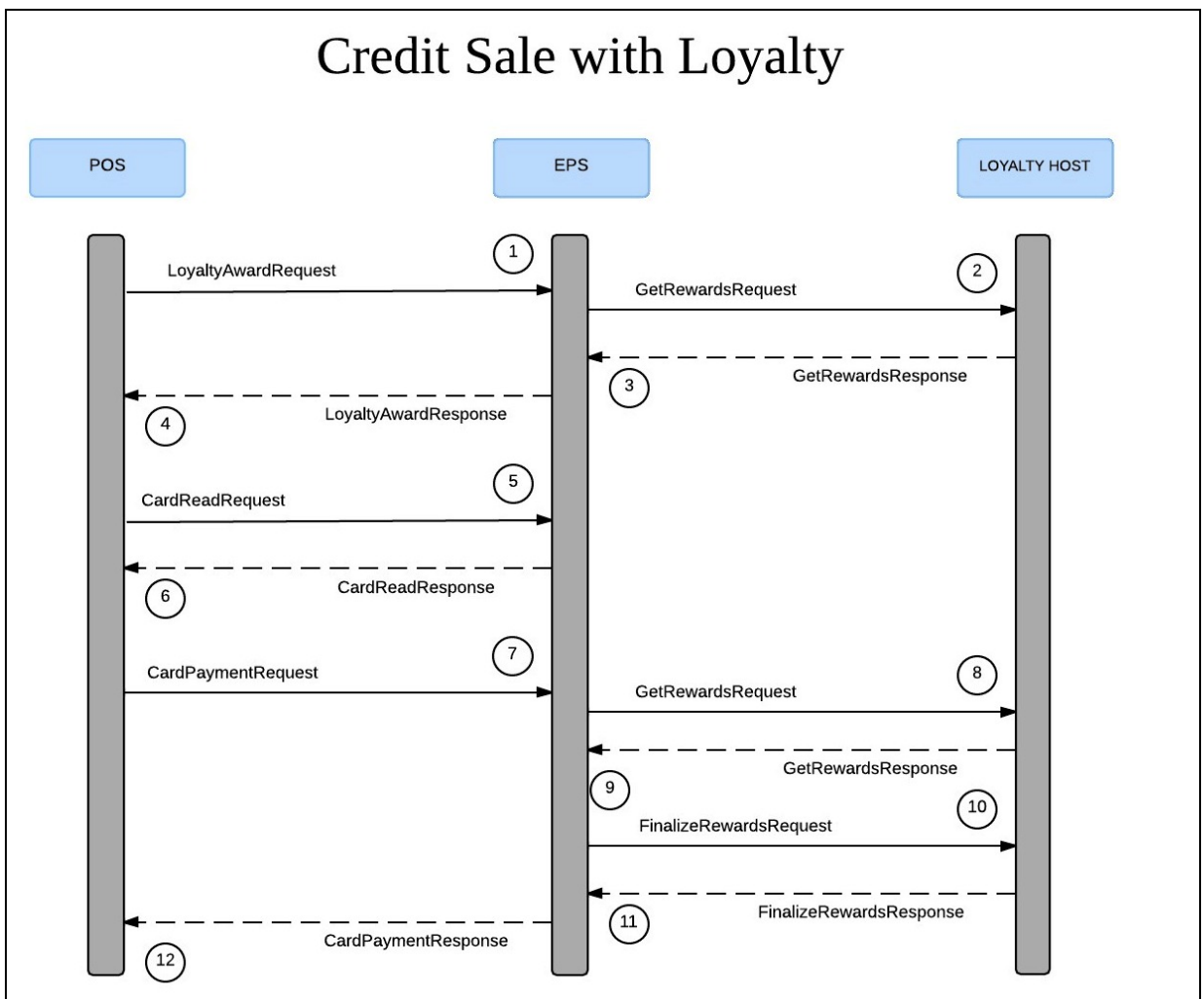
CancelTransaction is used to inform the Loyalty system that the customer or cashier has cancelled or voided a transaction in progress, before it was finalized. The loyalty system will then cancel or rollback any customer accruals or redemptions pending for the transaction at this point.

Reversal Transaction is used to inform the loyalty system that the cashier has cancelled, voided or refunded a previously finalized transaction. The Loyalty system will then cancel or rollback any customer accruals or redemptions processed for the original transaction. In the case of a ReversalTransaction, the EPS will send the previously finalized loyalty transaction details to the loyalty system that needs to be reversed.

3.5 Transaction Flow

1. Credit Sale with Loyalty.

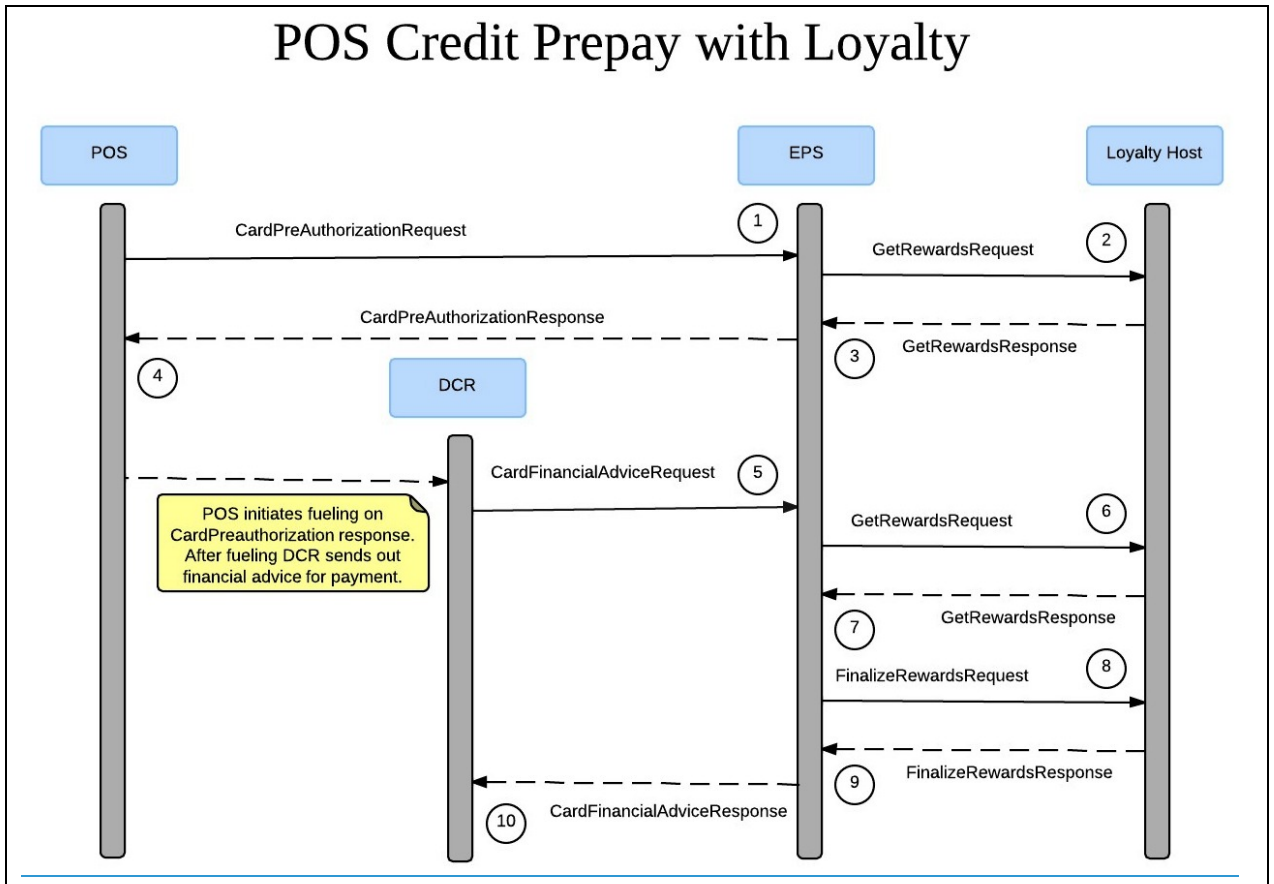
Description	This section describes the process flow involved in a normal credit transaction with loyalty.
Trigger	Cashier rings up merchandise items and presses 'Credit' key. We assume that the customer must have entered a Loyalty followed by Payment to get the discount. If the 'Auth on Total' is enabled for the transaction, the first LoyaltyAward request is sent on 'Total' key press, or else it will be sent on 'MOP' key.
	Note: Any merchandise (fuel/non-fuel) can be eligible for loyalty discount. Fuel items can get PPG discount, Transaction-Level discount, or Item-Level discount (line item discount), where non-fuel products can get Item-Level discount, which is entirely based on the loyalty host.



Note: With the inclusion of line item discounts, any non-fuel item can be eligible for loyalty discount. Therefore, this flow suits for any fuel/non-fuel loyalty transactions

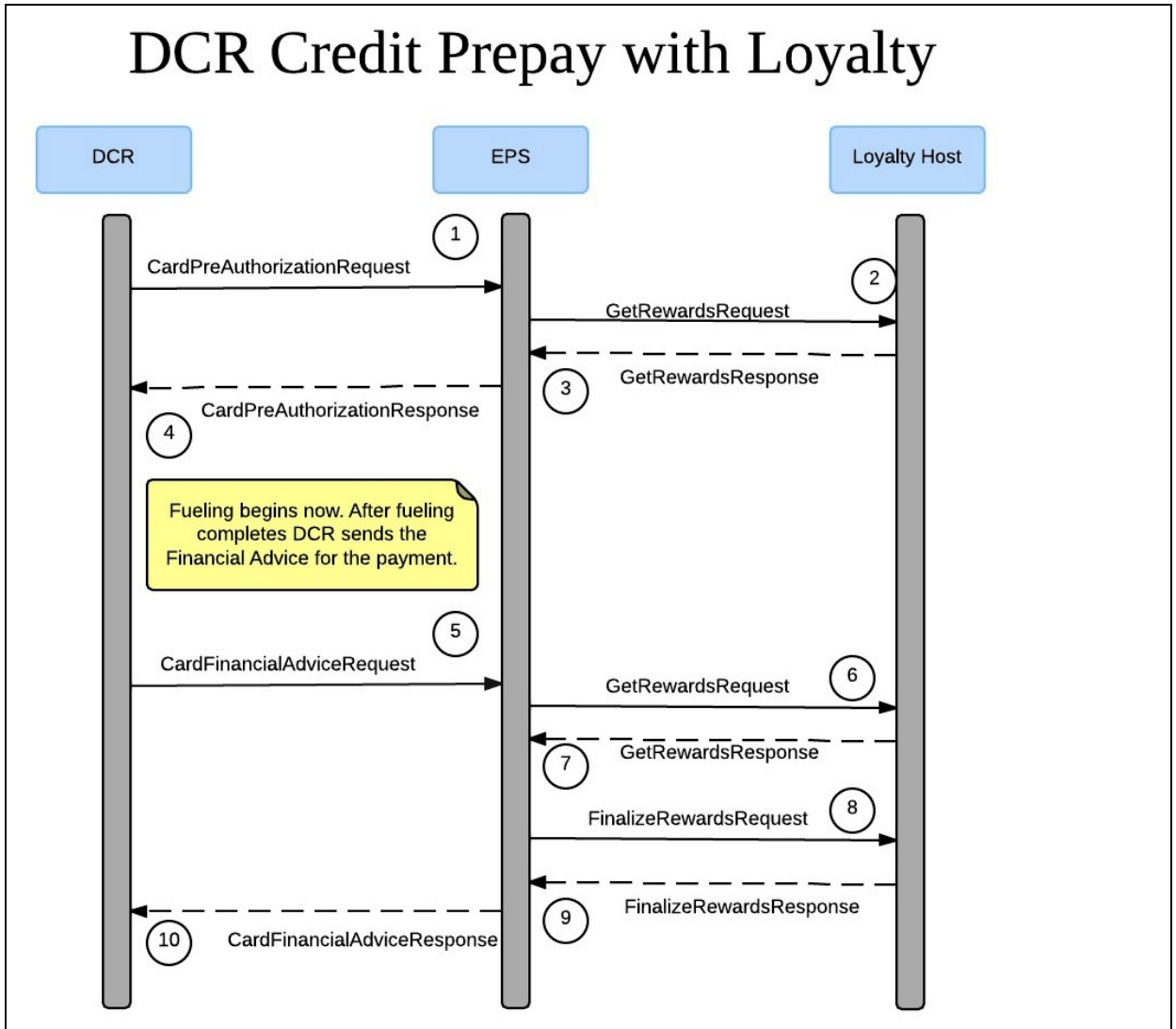
2. POS Credit Prepay with Loyalty.

Description	This section describes the process flow involved in a credit prepay transaction with loyalty on the POS.
Trigger	Cashier performs fuel prepay and presses 'Credit' key. We assume tht the customer must have swiped Loyalty ID followed by Payment to get the discount.



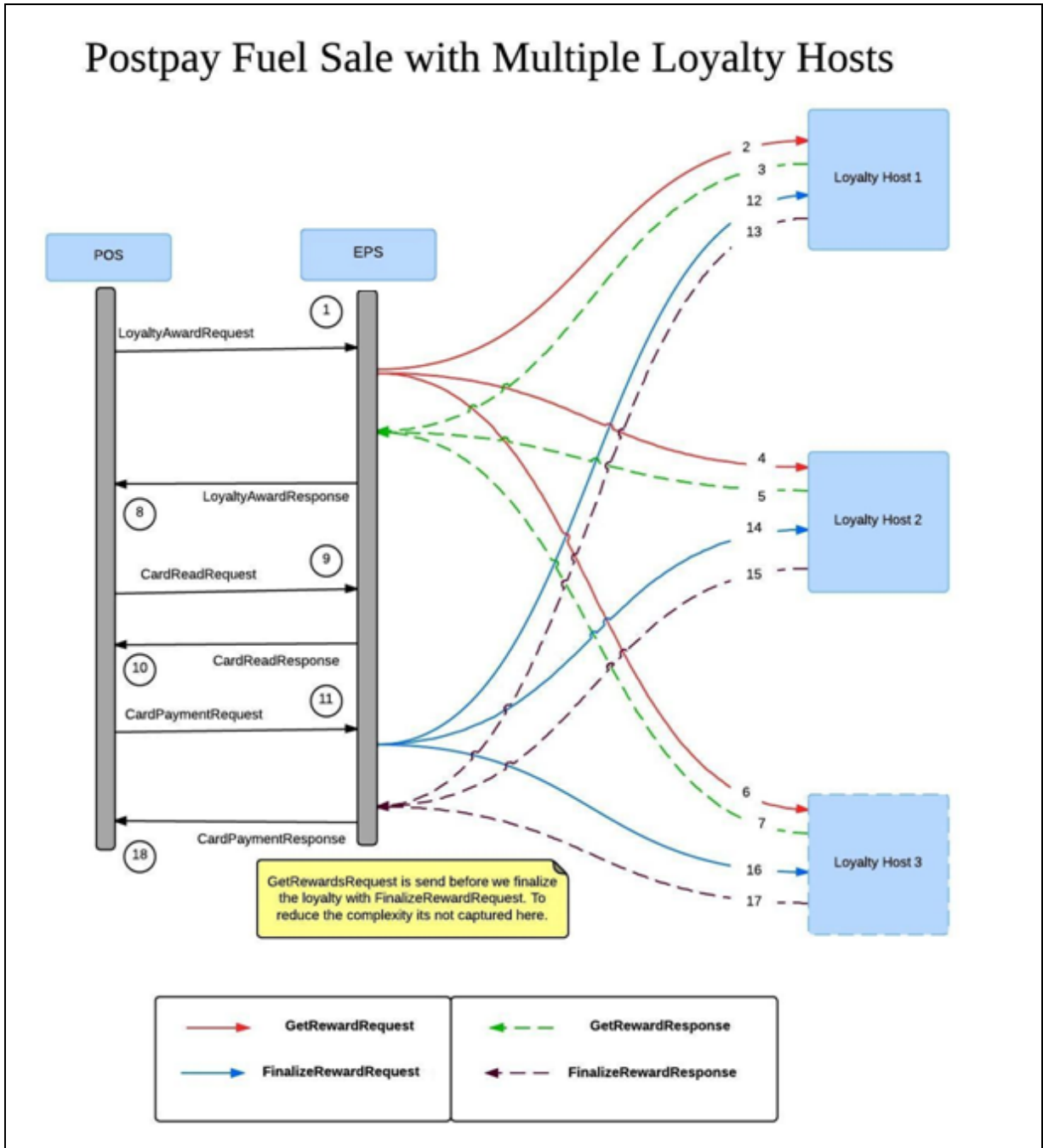
3. DCR Credit Prepay with Loyalty

Description	This section describes the process flow involved in a credit prepay transaction with loyalty on DCR.
Trigger	We assume that the customer must have swiped Loyalty card followed by Payment card at the DCR to initiate the fuel transaction with loyalty.



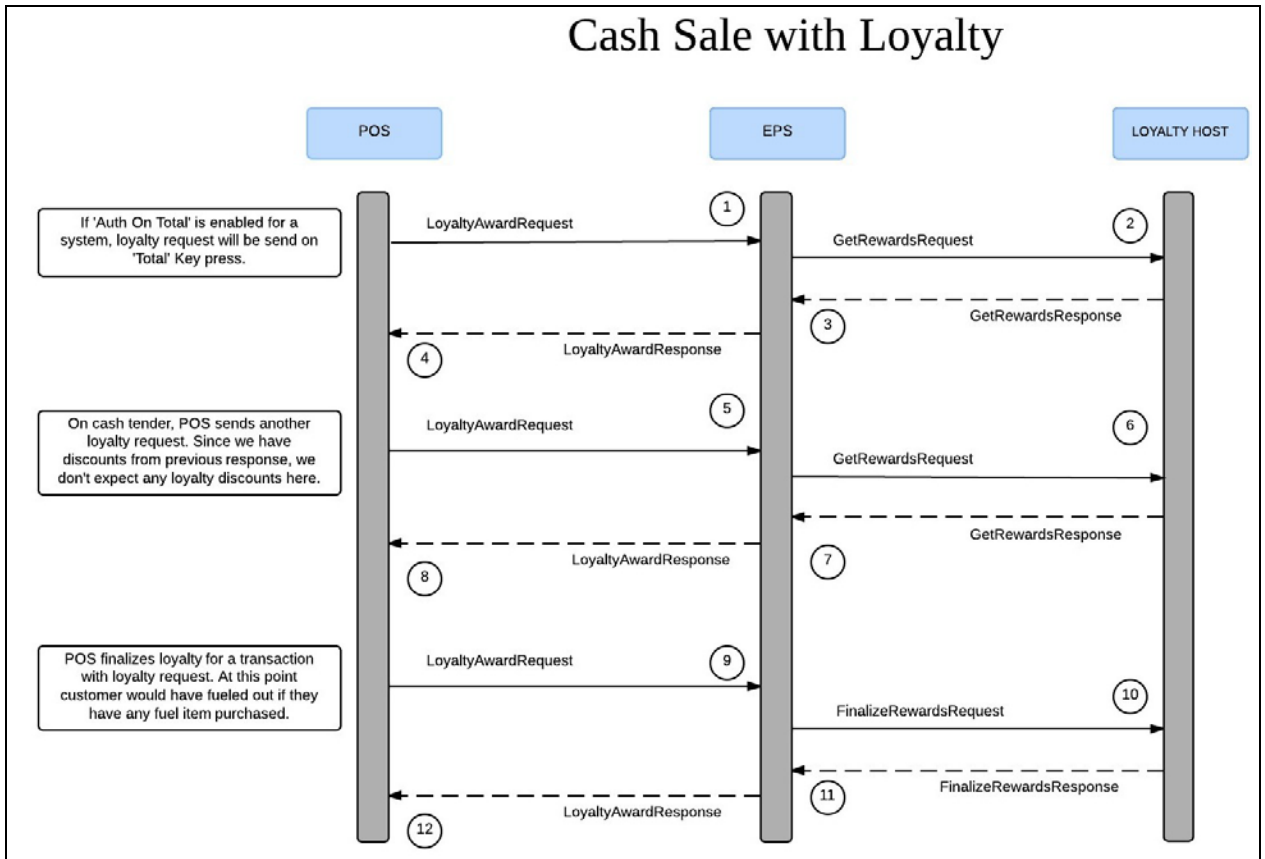
4. Multiple Loyalty Transaction Flow

Description	This section describes the process flow involved in a multiple loyalty system.
Trigger	Cashier rings up post fuel item and presses 'Credit' key. Customer must have swiped one or more Loyalty card with a Payment card to get multiple discounts from multiple loyalty hosts.



5. Cash Loyalty Transaction Flow

Description	This section describes the process flow involved in a loyalty transaction with cash tender.
Trigger	Cashier rings up prepay fuel item and some non -fuel sale items and presses 'Total' key followed by 'Cash' MOP key. We assume that customer/cashier must have entered a loyalty ID at this point.



EPS to Loyalty Host — XML Requests & Responses Sample:

GetRewardRequest XML

```
<ns2:GetRewardsRequest xmlns:ns2="http://www.pcats.org/schema/naxml/loyalty/v01"
xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.naxml.org/POSBO/Vocabulary/2003-10-16">
  <ns2:RequestHeader>
    <ns2:POS LoyaltyInterfaceVersion>X.X</ns2:POS LoyaltyInterfaceVersion>
    <ns3:VendorName>VIPER:Topaz</ns3:VendorName>
    <ns3:VendorModelVersion>vX.0X:X.00.00</ns3:VendorModelVersion>
    <ns2:POSSequenceID>XXXXXXXXXXXXXXXXXX</ns2:POSSequenceID>
    <ns2:StoreLocationID>XXXXXXXXXXXXXXXXXX</ns2:StoreLocationID>
    <ns2:LoyaltyOfflineFlag value="no"/>
  </ns2:RequestHeader>
  <ns2:LoyaltyID entryMethod="swipe-
track">XXXXXX*XXXX<ns2:Track2>XXXXXXXXXXXX</ns2:Track2>
  </ns2:LoyaltyID>
  <ns2:FuelPriceData/>
  <ns2:TransactionData>
    <ns2:TransactionHeader>
      <ns3:OutsideSalesFlag value="no"/>
      <ns3:RegisterID>XX</ns3:RegisterID>
      <ns3:CashierID>X</ns3:CashierID>
      <ns2:TillID>XXX</ns2:TillID>
      <ns2:POSTransactionID>XXXXXXXXXXXX</ns2:POSTransactionID>
      <ns2:BusinessPeriod>
        <ns3:BusinessDate>2013-01-31</ns3:BusinessDate>
        <ns3:PrimaryReportPeriod
interval="other">X</ns3:PrimaryReportPeriod>
        <ns3:BeginDate>2013-01-18</ns3:BeginDate>
        <ns3:BeginTime>05:12:20</ns3:BeginTime>
        <ns3:EndDate>2100-01-01</ns3:EndDate>
        <ns3:EndTime>00:00:00</ns3:EndTime>
      </ns2:BusinessPeriod>
      <ns3:EventStartDate>2013-01-31</ns3:EventStartDate>
      <ns3:EventStartTime>19:34:16</ns3:EventStartTime>
    </ns2:TransactionHeader>
    <ns2:TransactionDetailGroup>
      <ns2:TransactionLine status="normal">
        <ns2:LineNumber>1</ns2:LineNumber>
        <ns2:ItemLine discountable="yes">
          <ns3:Description>TESTPRODUCT1</ns3:Description>
          <ns3:ActualSalesPrice>XX.XX</ns3:ActualSalesPrice>
          <ns3:RegularSellPrice>XX.XX</ns3:RegularSellPrice>
          <ns3:SalesQuantity>X</ns3:SalesQuantity>
          <ns3:SalesAmount>XX.XX</ns3:SalesAmount>
          <ns2:SalesUOM>EA</ns2:SalesUOM>
        </ns2:ItemLine>
      </ns2:TransactionLine>
      <ns2:TransactionLine status="normal">
        <ns2:LineNumber>2</ns2:LineNumber>
        <ns2:ItemLine discountable="yes">
```

```

<ns3:Description>TESTPRODUCT2</ns3:Description>
<ns3:ActualSalesPrice>XX.XX</ns3:ActualSalesPrice>
<ns3:RegularSellPrice>XX.XX</ns3:RegularSellPrice>
<ns3:SalesQuantity>X</ns3:SalesQuantity>
<ns3:SalesAmount>XX.XX</ns3:SalesAmount>
<ns2:SalesUOM>EA</ns2:SalesUOM>

<ns3:PaymentSystemsProductCode>XXX</ns3:PaymentSystemsProductCode>
</ns2:ItemLine>
</ns2:TransactionLine>
</ns2:TransactionDetailGroup>
</ns2:TransactionData>
</ns2:GetRewardsRequest>

```

GetRewardsResponse XML

```

<ns3:GetRewardsResponse xmlns:ns2="http://www.naxml.org/POSBO/Vocabulary/2003-10-16"
xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.pcats.org/schema/naxml/loyalty/v01">
  <ns3:ResponseHeader overallResult="success">
    <ns3:POS LoyaltyInterfaceVersion>X.X</ns3:POS LoyaltyInterfaceVersion>
    <ns2:VendorName>VIPER:Topaz</ns2:VendorName>
    <ns2:VendorModelVersion>vX.0X:X.00.00</ns2:VendorModelVersion>
    <ns3:POSSequenceID>XXXXXXXXXXXXXXXXXX</ns3:POSSequenceID>
    <ns3:LoyaltySequenceID>XX</ns3:LoyaltySequenceID>
    <ns4:Result>
      <Success/>
    </ns4:Result>
  </ns3:ResponseHeader>
  <ns3:LoyaltyIDValidFlag value="yes">XXXXXXXXXXXX</ns3:LoyaltyIDValidFlag>
  <ns3:RewardActions>
    <ns3:AddReward>
      <ns3:LoyaltyRewardID>Multi</ns3:LoyaltyRewardID>
      <ns3:InstantRewardFlag value="yes"/>
      <ns3:RewardTargetLineNumber>2</ns3:RewardTargetLineNumber>
      <ns3:RewardDiscountMethod>amountOff</ns3:RewardDiscountMethod>
      <ns3:RewardValue>XX.XX</ns3:RewardValue>
      <ns3:RewardLimit>20</ns3:RewardLimit>
      <ns3:RewardReceiptDescShort>DISCOUNT</ns3:RewardReceiptDescShort>
    </ns3:AddReward>
  </ns3:RewardActions>
  <ns3:CustomerMessageData>
    <ns3:DisplayData>
      <ns3:DisplayCommand duration="3" sequence="WhenReceived"
device="OPT">
        <ns3:DisplayLine>CUSTOMER DISPLAY
MESSAGE</ns3:DisplayLine>
      </ns3:DisplayCommand>
    </ns3:DisplayData>
  </ns3:CustomerMessageData>
</ns3:GetRewardsResponse>

```

FinalizeRewardsRequest XML

```
<ns2:FinalizeRewardsRequest xmlns:ns2="http://www.pcats.org/schema/naxml/loyalty/v01"
xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.naxml.org/POSBO/Vocabulary/2003-10-16">
  <ns2:RequestHeader>
    <ns2:POS LoyaltyInterfaceVersion>X.X</ns2:POS LoyaltyInterfaceVersion>
    <ns3:VendorName>VIPER:Topaz</ns3:VendorName>
    <ns3:VendorModelVersion>vX.0X:X.00.00</ns3:VendorModelVersion>
    <ns2:POSSequenceID>XXXXXXXXXXXXXXXX</ns2:POSSequenceID>
    <ns2:LoyaltySequenceID>XX</ns2:LoyaltySequenceID>
    <ns2:StoreLocationID>XXXXXXXXXXXXXXXX</ns2:StoreLocationID>
    <ns2:LoyaltyOfflineFlag value="no"/>
  </ns2:RequestHeader>
  <ns2:LoyaltyID entryMethod="swipe-
track">XXXXXXXX*XXXX<ns2:Track2>XXXXXXXXXXXXXXXX</ns2:Track2>
  </ns2:LoyaltyID>
  <ns2:TransactionData>
    <ns2:TransactionHeader>
      <ns3:OutsideSalesFlag value="no"/>
      <ns3:RegisterID>XX</ns3:RegisterID>
      <ns3:CashierID>XX</ns3:CashierID>
      <ns2:TillID>011</ns2:TillID>
      <ns2:POSTransactionID>XXXXXXXXXX</ns2:POSTransactionID>
      <ns2:BusinessPeriod>
        <ns3:BusinessDate>2013-01-31</ns3:BusinessDate>
        <ns3:PrimaryReportPeriod
interval="other">1</ns3:PrimaryReportPeriod>
        <ns3:BeginDate>2013-01-18</ns3:BeginDate>
        <ns3:BeginTime>05:12:20</ns3:BeginTime>
        <ns3:EndDate>2100-01-01</ns3:EndDate>
        <ns3:EndTime>00:00:00</ns3:EndTime>
      </ns2:BusinessPeriod>
      <ns3:EventStartDate>2013-01-31</ns3:EventStartDate>
      <ns3:EventStartTime>19:34:16</ns3:EventStartTime>
    </ns2:TransactionHeader>
    <ns2:TransactionDetailGroup>
      <ns2:TransactionLine status="normal">
        <ns2:LineNumber>1</ns2:LineNumber>
        <ns2:ItemLine discountable="yes">
          <ns3:Description>TESTPRODUCT1</ns3:Description>
          <ns3:ActualSalesPrice>XX.XX</ns3:ActualSalesPrice>
          <ns3:RegularSellPrice>XX.XX</ns3:RegularSellPrice>
          <ns3:SalesQuantity>XX</ns3:SalesQuantity>
          <ns3:SalesAmount>XX.XX</ns3:SalesAmount>
          <ns2:SalesUOM>EA</ns2:SalesUOM>
        </ns2:ItemLine>
      </ns2:TransactionLine>
      <ns2:TransactionLine status="normal">
        <ns2:LineNumber>2</ns2:LineNumber>
        <ns2:ItemLine discountable="yes">
          <ns3:Description>TESTPRODUCT2</ns3:Description>
```

```

        <ns3:ActualSalesPrice>XX.XX</ns3:ActualSalesPrice>
        <ns3:RegularSellPrice>XX.XX</ns3:RegularSellPrice>
        <ns3:SalesQuantity>XX</ns3:SalesQuantity>
        <ns3:SalesAmount>XX.XX</ns3:SalesAmount>
        <ns2:SalesUOM>EA</ns2:SalesUOM>
        <ns2:Promotion>
            <ns2:PromotionID>Multi</ns2:PromotionID>
            <ns2:LoyaltyRewardID>Multi</ns2:LoyaltyRewardID>

        <ns2:PromotionAmount>XX.XX</ns2:PromotionAmount>

        <ns3:PromotionReason>loyaltyOffer</ns3:PromotionReason>
        </ns2:Promotion>

        <ns3:PaymentSystemsProductCode>XXX</ns3:PaymentSystemsProductCode>
        </ns2:ItemLine>
        </ns2:TransactionLine>
        <ns2:TransactionLine status="normal">
            <ns2:LineNumber>3</ns2:LineNumber>
            <ns2:TenderInfo>
                <ns3:TenderCode>creditCards</ns3:TenderCode>

                <ns2:TenderSubCode>MASTERCARD</ns2:TenderSubCode>
                <ns2:ISOPrefix>XXXXXX</ns2:ISOPrefix>
                <ns3:TenderAmount
currency="USD">XX.XX</ns3:TenderAmount>
                <ns3:ChangeFlag value="no"/>
            </ns2:TenderInfo>
        </ns2:TransactionLine>
        </ns2:TransactionDetailGroup>
    </ns2:TransactionData>
</ns2:FinalizeRewardsRequest>

```

FinalizeRewardResponse XML

```

<ns3:FinalizeRewardsResponse xmlns:ns2="http://www.naxml.org/POSBO/Vocabulary/2003-10-16"
xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.pcats.org/schema/naxml/loyalty/v01">
    <ns3:ResponseHeader overallResult="success">
        <ns3:POS LoyaltyInterfaceVersion>X.X</ns3:POS LoyaltyInterfaceVersion>
        <ns2:VendorName>VIPER:Topaz</ns2:VendorName>
        <ns2:VendorModelVersion>vX.0X:X.00.00</ns2:VendorModelVersion>
        <ns3:POSSequenceID>XXXXXXXXXXXXXXXXXX</ns3:POSSequenceID>
        <ns3:LoyaltySequenceID>XX</ns3:LoyaltySequenceID>
        <ns4:Result>
            <Success/>
        </ns4:Result>
    </ns3:ResponseHeader>
    <ns3:CustomerMessageData>
        <ns3:DisplayData>
            <ns3:DisplayCommand duration="3" sequence="WhenReceived"
device="OPT">
                <ns3:DisplayLine>CUSTOMER DISPLAY
MESSAGE</ns3:DisplayLine>

```

```

        </ns3:DisplayCommand>
        </ns3:DisplayData>
        <ns3:ReceiptData>
            <ns3:ReceiptLine>RECEIPT TEXT</ns3:ReceiptLine>
        </ns3:ReceiptData>
        </ns3:CustomerMessageData>
    </ns3:FinalizeRewardsResponse>

```

GetLoyaltyOnlineStatusRequest XML

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns3:GetLoyaltyOnlineStatusRequest
  xmlns:ns2="http://www.naxml.org/POSBO/Vocabulary/2003-10-16"
  xmlns:ns4="http://www.pcats.org/schema/core/v01"
  xmlns:ns3="http://www.pcats.org/schema/naxml/loyalty/v01">
  <ns3:RequestHeader>
    <ns3:POS LoyaltyInterfaceVersion>1.0</ns3:POS LoyaltyInterfaceVersion>
    <ns2:VendorName>VIPER</ns2:VendorName>
    <ns2:VendorModelVersion>5.07.00</ns2:VendorModelVersion>
    <ns3:POSSequenceID>XXX</ns3:POSSequenceID>
    <ns3:LoyaltySequenceID/>
    <ns3:StoreLocationID>XXXXXXXXXXXX</ns3:StoreLocationID>
    <ns3:LoyaltyOfflineFlag value="no"/>
  </ns3:RequestHeader>
</ns3:GetLoyaltyOnlineStatusRequest>

```

GetLoyaltyOnlineStatusResponse XML

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns3:GetLoyaltyOnlineStatusResponse
  xmlns:ns2="http://www.naxml.org/POSBO/Vocabulary/2003-10-16"
  xmlns:ns4="http://www.pcats.org/schema/core/v01"
  xmlns:ns3="http://www.pcats.org/schema/naxml/loyalty/v01">
  <ns3:ResponseHeader overallResult="success">
    <ns3:POS LoyaltyInterfaceVersion>1.0</ns3:POS LoyaltyInterfaceVersion>
    <ns2:VendorName>VIPER</ns2:VendorName>
    <ns2:VendorModelVersion>5.07.00</ns2:VendorModelVersion>
    <ns3:POSSequenceID> XXX </ns3:POSSequenceID>
    <ns3:LoyaltySequenceID>XX</ns3:LoyaltySequenceID>
    <ns4:Result>
      <Success/>
    </ns4:Result>
  </ns3:ResponseHeader>
  <ns3:PromptForLoyaltyFlag value="yes"/>
</ns3:GetLoyaltyOnlineStatusResponse>

```

CancelRedemptionMessage XML

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns3:CancelRedemptionRequest xmlns:ns2="http://www.naxml.org/POSBO/Vocabulary/2003-10-16" xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.pcats.org/schema/naxml/loyalty/v01">
  <ns3:RequestHeader>
    <ns3:POS LoyaltyInterfaceVersion>1.0</ns3:POS LoyaltyInterfaceVersion>
    <ns2:VendorName>VIPER:Topaz</ns2:VendorName>
    <ns2:VendorModelVersion>5.07.00:1.00.00</ns2:VendorModelVersion>
    <ns3:POSSequenceID> XXXXXXXXXXXX </ns3:POSSequenceID>
    <ns3:LoyaltySequenceID> XX </ns3:LoyaltySequenceID>
    <ns3:StoreLocationID> XXXXXXXXXXXX </ns3:StoreLocationID>
  </ns3:RequestHeader>
  <ns3:LoyaltyRewardID>RewardID1</ns3:LoyaltyRewardID>
  <ns3:LoyaltyRewardID>RewardID2</ns3:LoyaltyRewardID>
</ns3:CancelRedemptionRequest>
```

3.6 Discount Types

The EPS Loyalty FEP module will support the following loyalty discount types.

PPG (Price Per Gallon)

The EPS Loyalty FEP module will receive PPG discount amount and quantity limit from Loyalty Host as a child element (FuelGradeReward) of the PrepayFuelRewards . This data will be sent to the POS as a child element of ‘SaleItem’ in the Response data.

GetRewardsResponse Message Details

Element	Type (Mandatory/Optional/Child)	Description
PrepayFuelRewards	O	
RewardDiscountMethod	M-C	Used to calculate a discount to be applied when redeeming a reward. This method can take "amountOffPPU" values. EPS will ignore the discount if any other values are supplied (eg; newPrice, percentOff)
RewardValue	O-C	The value should not be greater than the actual price of the product (price will not roll back below minimum price [\$]).In such scenario, EPS will ignore the discount.
InstantRewardFlag	M-C	The attribute value ="no" implies optional discount.

Transaction Level (Ticket - Level)

In addition to the normal transaction-level discount scenarios where the cashier/customer opts for the discount at the end of the transaction, transaction-level discounts can also be given to the customer when POS is unable to roll back the price at the pump — even after POS got the PPG amount from the EPS and the customer has accepted to apply PPG discount.

In such scenario, the POS will send the CardFinancialAdvice request to the EPS with the price adjustment amount is zero. This 'amount' will be represented under the 'SaleItem' element. EPS will then send the 'GetRewardsRequest' message to Loyalty Host with the value 'zero' in the 'Promotion' element inside the 'TransactionData' segment. The Loyalty Host will now send the transaction-level discount amount in the 'GetRewardsResponse' message to the EPS so that the EPS can send that amount in the response to the POS. Since the customer has already accepted to go for PPG discount, this will be applied to the transaction as 'Mandatory Discount'.

GetRewardsRequest Message Details

Element	Type (Mandatory/Optional/Child)	Description
TransactionData	O	The transaction details received from the POS.
TransactionDetailGroup	O-C	Contains one or more transaction lines.
TransactionLine	M-C	Represents a single line of transaction.
FuelLine	M-C	Data elements that describe a fuel sale item.
Promotion	O-C	The promotional discount applied automatically to the fuel.
PromotionAmount	M-C	The loyalty discount amount already applied to the transaction. Note: In case PPG discount could not be applied, this will have zero as the amount.

GetRewardsResponse Message Details

Element	Type (Mandatory/Optional/Child)	Description
AddReward	O	
RewardDiscountMethod	M-C	Used to calculate a discount to be applied when redeeming a reward. This element can take as a value "amountOff" only. EPS will ignore the discount if any other values are supplied (eg; newPrice, percentOff)
RewardValue	O-C	The value should not be greater than the total transaction amount. [EPS will ignore the discount if the discount amount is greater than the total transaction amount].
InstantRewardFlag	M-C	The attribute value ="no" implies optional discount and "yes" implies mandatory discount.

Item Level (Line Item Discount)

A loyalty host can provide discount for a PLU line item in a transaction. These discounts can be instant or optional. If instant, discounts are added to the transaction immediately on getting the discounts. In case of optional discounts, a customer is prompted for an approval to apply that discount. If there are ***n*** items which are eligible for optional discounts, then there will be ***n*** number of approval prompts.

On receiving the discount from the loyalty host, a new price adjustment element is added to the PLU item. All the details for the discount are added to that price adjustment element except unit price, since unit price is considered only for PPG discount. If there is a price adjustment element attached to the PLU item while applying line-item discounts, then the discount amount is updated with the new discount value, assuming that a loyalty host will have a reason to change the value of already applied PLU item discount.

The POS can set the "PriceChangeEligible" attribute to false on any sale item (by default). For any PLU (MerchandiseCodeLine) or department (ItemLine) item. The EPS loyalty FEP will set the "discountable" flag as false and the Loyalty host should not apply a discount on this item.

The POS can also set a fuel prepay item as ineligible (for example, when the POS has already applied a local discount with higher precedence and it is not possible to stack PPG discount). In such cases, the EPS loyalty FEP will not include the fuel price data as part of the GetRewards request. It will include only the transaction data. The Loyalty host can still give the item-level or transaction-level discount.

GetRewardsRequest Message Details

Element	Type (Mandatory/Optional/Child)	Description
TransactionData	O	The transaction details received from POS.
TransactionDetailGroup	O-C	Contains one or more transaction lines.
TransactionLine	M-C	Represents a single line of transaction.
ItemLine	M-C	Data elements that describe a PLU item line.
Promotion	O-C	The promotional discount applied automatically to the fuel.
PromotionAmount	M-C	The loyalty discount amount already applied to the transaction. Note: In case PPG discount could not be applied, this will have zero as the amount.
MerchandisecodeLine	M-C	Data elements that describe a department sale.
Promotion	O-C	The promotional discount applied automatically to the fuel.
PromotionAmount	M-C	The loyalty discount amount already applied to the transaction. Note: In case PPG discount could not be applied, this will have zero as the amount.

GetRewardsResponse Message Details

Element	Type (Mandatory/Optional/Child)	Description
AddReward	O	
LoyaltyRewardID	M-C	Generated by the loyalty system and associated with corresponding POS promotion as a unique identifier.
InstantRewardFlag	M-C	The attribute value ="no" implies optional discount and "yes" implies mandatory discount.
RewardTargetLineNumber	M-C	Used to indicate the sale item's line number to which the discount is to be applied.
RewardDiscountMethod	M-C	Used to calculate a discount to be applied when redeeming a reward. This element can take as a value "amountOff" only. EPS will ignore the discount if any other values are supplied (eg; newPrice, percentOff)
RewardValue	O-C	The value should not be greater than the total transaction amount. [EPS will ignore the discount if the discount amount is greater than the total transaction amount].

Mandatory Discount

Customer would not be prompted for the discount amount if the FEP Loyalty Host sends a mandatory discount.

GetRewardsResponse Message

Element	Type (Mandatory/Optional/Child)	Description
PrepayFuelRewards/AddReward	O	
InstantRewardFlag	M	The attribute value = "yes" implies the discount is a mandatory (forced).

3.7 Receipt Messages

Receipt messages are obtained as part of the 'FinalizeRewardsResponse' message received from the Loyalty host and are sent to POS to be printed at the end of the receipt. The EPS Loyalty FEP module will not send receipt messages to the EPS if the Loyalty Host is not available for receipt messages.

Long and Short Receipts

For indoor transactions, the long indoor receipt (RewardReceiptDescLong) is used, if available; otherwise the short receipt (RewardReceiptDescShort) is used. The short receipt is always used for outdoor transactions.

3.8 Prompt Messages

Prompt messages can be displayed indoors at the PINpad or outside at the payment terminal.

Long and Short Prompts

Prompt texts are applicable only if "InstantRewardFlag" is set to NO, which implies it is an optional discount. The prompt text elaborates what needs to be displayed to the customer while seeking a YES/NO answer. For indoor workstations, the long prompt text (RewardPromptTextLong) is used, if available; otherwise the short prompt text (RewardPromptTextShort) is used. The short prompt text is always used for outdoor (DCR) workstations.

3.9 Customer Messages

Customer messages are obtained from the loyalty host as part of the loyalty host response. When received, the EPS sends display prompts to the logged-in customer or cashier devices based on the target device specified by the host in the response. Customer message display is supported as part of 'GetRewardsResponse' on any type of discounts (Instant or Optional) and for the "FinalizeRewardsResponse". It is recommended to avoid including customer messages in order to keep processing times to a minimum for prepay fuel and outdoor transactions. The recommended approach is to include this information in the receipt data for the "FinalizeRewardsResponse".

Text Limitation

The EPS will be sending the display message requests for target devices mentioned in the 'DisplayCommand' field of the 'CustomerMessageData' element present in 'GetRewardsResponse'. Currently indoor devices can

support up to 3 lines, each with length of 20 characters and outdoor devices can display a maximum of 2 lines, each with a maximum length of 16 characters. Since the schema only provides for a single line of text for the customer messages, the EPS will format the message for display based on the target device.

3.10 Cash Prepay – Double Finalize

In case of a cash prepay, EPS may be sending two FinalizeReward message to the loyalty host system.

Note: Double finalization happens on cash prepay only in Ruby, Ruby/Sapphire, or Topaz/Sapphire environments. The Commander Site Controller does not send two FinalizeReward messages.

For cash prepay with or without merchandise, the EPS will send a 'Finalize Reward' message to the loyalty host system once a cash payment happens. It does complete all the merchandise, merchandise discount, and MOP's up to that point.

But the first 'FinalizeReward' message will not finalize loyalty on the fuel item present. Since it is a prepay item, it will be finalized only after the customer fuels out. Hence, the fuel portion is completed with possible additional 'GetReward' messages and a separate, but linked, FinalizeReward' message for the dispensed fuel.

3.11 Fallback Processing

When the Loyalty Host is offline, the EPS Loyalty FEP Module will approve the loyalty transaction requests coming from the POS which result in creating 'GetRewardsRequest' messages. Approving the loyalty transactions will not result in applying loyalty discounts. This mechanism will use Loyalty Action Code Table to approve the transactions if the Loyalty card match is found.

When the loyalty host is offline while sending the 'GetRewardRequest', the loyalty FEP module will approve the transaction without a loyalty discount. When the host comes back online, the loyalty earned information will be sent to the host.

The Loyalty FEP Module cannot approve all other loyalty transaction requests. The Loyalty FEP Module, then, will send a response message to the POS indicating that the loyalty host is not available. This message will be retrieved from the text table.

Some of the loyalty transaction requests from POS, which result in certain request messages to be sent to the Loyalty Host, will be processed in the "Store and Forward" method during fallback. Those messages and the processing of those messages are explained in the "Store and Forward" section.

3.12 Store and Forward

The Store and Forward mechanism will enable the EPS to store transactions when the Loyalty Host is offline and send them to the Loyalty Host once the Loyalty Host is online.

If the loyalty host is offline while sending the following requests, the corresponding message will be stored in the batch item as uncollected.

- FinalizeRewards

- CancelTransactions
- ReverseTransactions
- EndPeriod

The uncollected batch items will be sent to the loyalty host once the loyalty host comes online.

There is maximum limit to the number of offline messages that can be stored in the store and forward batch item as uncollected. This maximum limit is controlled by the FEPTable attribute 'StoreAndForwardMaximum'. When it reaches to this maximum limit, all further requests will not get authorized offline.

The following table shows the store and forward mechanism used for the message types that can utilize this mechanism.

The EPS will use "GetLoyaltyOnlineStatus" message to find the Loyalty Host Online status.

Note: The 'Host is Online' scenario described below implies the loyalty host becomes online from the offline state.

Message to be sent to Loyalty Host	Scenario	Store and Forward Mechanism	Response to POS
FinalizeRewards	Host is offline	The message will be stored by Loyalty FEP Module with the 'LoyaltyOfflineFlag' set to 'yes'.	Response message indicating that the Loyalty Host is offline will be sent.
FinalizeRewards	Host is online	The stored message will be submitted to the Host.	N/A
CancelTransaction	Host is offline	The message will be stored by Loyalty FEP Module with the 'LoyaltyOfflineFlag' set to 'yes'	N/A
CancelTransaction	Host is online	The stored message will be submitted to the Host.	N/A
ReverseTransaction	Host is offline	The message will be stored by Loyalty FEP Module with the 'LoyaltyOfflineFlag' set to 'yes'.	Response message indicating that the Loyalty Host is offline will be sent.
ReverseTransaction	Host is Online	The stored message will be submitted to the Host.	N/A
EndPeriod	Host is Offline	The message will be stored by Loyalty FEP	Response message indicating that the

		Module with the 'LoyaltyOfflineFlag' to 'yes'.	Loyalty Host is offline will be sent.
EndPeriod	Host is Online	The stored message will be submitted to the Host.	N/A

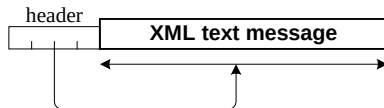
In the case of Store and Forward, if the 'OverallResult' attribute of 'ResponseHeader' element in Loyalty Host response has a 'failure' value, then the request will be logged.

3.13 Header Definition

VeriFone implements a single LFEP header definition and this must be supported by any partner.

The currently supported LHFEP module was adopted from the PCATS/IFSF POS to EPS messaging standard and is four bytes long.

The TCP data transport is a stream of service. In order to detect the end of the XML message, a header of four bytes containing the length of the following XML request/response message is added before the XML message. The size in the header will be in Big Endian format. All the loyalty messages, both request and response, will be preceded by this header indicating the size of the following XML message. The following diagram shows the header followed by the XML text message.



Example

For the following XML text message (Part of GetRewardsRequest XML message)

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns2:GetRewardsRequest
xmlns:ns2="http://www.pcats.org/schema/naxml/loyalty/v01"
xmlns:ns4="http://www.pcats.org/schema/core/v01"
xmlns:ns3="http://www.naxml.org/POSBO/Vocabulary/2003-10-16">
</ns2:GetRewardsRequest>
```

The "00 00 01 15" header is added proceeding the following data transferred inside the TCP connection.

0x000	0000 0115 3C3F 786D 6C20 7665 7273 696F	<?xml versio
0x010	6E3D 2231 2E30 2220 656E 636F 6469 6E67	n="1.0" encoding
0x020	3D22 5554 462D 3822 2073 7461 6E64 616C	="UTF-8" standal
0x030	6F6E 653D 2279 6573 223F 3E0D 0A3C 6E73	one="yes"?>..<ns
0x040	323A 4765 7452 6577 6172 6473 5265 7175	2:GetRewardsRequ
0x050	6573 7420 786D 6C6E 733A 6E73 323D 2268	est xmlns:ns2="h
0x060	7474 703A 2F2F 7777 772E 7063 6174 732E	ttp://www.pcats.
0x070	6F72 672F 7363 6865 6D61 2F6E 6178 6D6C	org/schema/naxml
0x080	2F6C 6F79 616C 7479 2F76 3031 2220 786D	/loyalty/v01" xm
0x090	6C6E 733A 6E73 343D 2268 7474 703A 2F2F	lns:ns4="http://
0x0A0	7777 772E 7063 6174 732E 6F72 672F 7363	www.pcats.org/sc
0x0B0	6865 6D61 2F63 6F72 652F 7630 3122 2078	hema/core/v01" x
0x0C0	6D6C 6E73 3A6E 7333 3D22 6874 7470 3A2F	mlns:ns3="http:/
0x0D0	2F77 7777 2E6E 6178 6D6C 2E6F 7267 2F50	/www.naxml.org/P
0x0E0	4F53 424F 2F56 6F63 6162 756C 6172 792F	OSBO/Vocabulary/
0x0F0	3230 3033 2D31 302D 3136 223E 0D0A 3C2F	2003-10-16">..</
0x100	6E73 323A 4765 7452 6577 6172 6473 5265	ns2:GetRewardsRe
0x110	7175 6573 743E	quest>

3.14 Action Codes

Each LHFEF can/will be using its own action codes and action code tables. The purpose of this action code table is to map various Host responses to a code/prompt for the POS. Decline and failure messages will be mapped through this action code table.

Action code tables can be shared among multiple hosts as long as it is agreed that the prompts will be same for each of the hosts. However, it is always preferred to have a separate action code table per host.

3.15 Receipts – Loyalty Portion

The EPS will populate only the loyalty portion of the receipt after the transaction has been finalized with the corresponding loyalty host.

Receipt example for prepay with loyalty

PREAUTH RECEIPT WITH LOYALTY		
WELCOME TO OUR STORE VeriFone Gold Disk		
FL		
Description	Qty	Amount

<CUSTOMER COPY>		
PREPAY CA #01		10.00

Subtotal		10.00
Tax		0.00
TOTAL		10.00
PREAUTH \$		10.00
PREPAY Receipt		
MASTERCARD \$10.00		
Acct/Card #: XXXXXXXXXXXX1491		
Auth #: 987652		
Resp Code: 2		
Stan: 0001120		
Invoice #: 7		
SITE ID: VP13007411001		
CUSTOMER COPY		
LINE 1 MESSAGE 1		
Card Balance: Unknown		
Last line		
REWARD		
Acct/Card #: XXXXXXXXXXXXX6197		
Trace #: 4		
Stan: 0001120		
ST# AB123 TILL XXXX DR# 1 TRAN# 1010022		
CSH: 1 07/19/13 17:52:25		

PREPAY FINAL RECEIPT WITH LOYALTY

**WELCOME TO
OUR STORE**

VeriFone Gold Disk

FL

Description	Qty	Amount
-----	---	-----

<CUSTOMER COPY>

UNLD1 CR #01	11.481G	10.00
--------------	---------	-------

SELF @ 0.871/ G

Subtotal	10.00
----------	-------

Tax	0.00
-----	------

TOTAL 10.00

CREDIT \$ 10.00

MASTERCARD

XXXXXXXXXXXX1491

Auth #: 987652

Resp Code: 0

Stan: 0001123

Invoice #: 7

SITE ID: VP13007411001

REWARD

XXXXXXXXXXXX6197

Trace #: 4

Stan: 0001123

Hi There

Welcome To our store again

Thank You

ST# AB123 TILL XXXX DR# 1 TRAN# 9010004

CSH: 0 07/19/13 17:54:29

Receipt example for transaction with ticket level discount

RECEIPT WITH TICKET LEVEL DISCOUNT		
WELCOME TO OUR STORE VeriFone Gold Disk		
FL		
Description	Qty	Amount

<CUSTOMER COPY>		
UNLD1 CA #01	12.6116	14.01
SELF @ 1.111/ G		
LINE ITEM DISC		-8.75

Subtotal		5.26
Tax		0.00
DISCOUNT		-3.25
TOTAL		2.01
CREDIT \$		2.01
MASTERCARD \$2.01 Acct/Card #: XXXXXXXXXXXX1491 Auth #: 987650 Resp Code: 0 Stan: 0001130 Invoice #: 8 SITE ID: VP13007411001 CUSTOMER COPY		
LINE 1 MESSAGE 1 Card Balance: Unknown Last line		
REWARD Acct/Card #: XXXXXXXXXXXX6197 Trace #: 5 Stan: 0001130 Hi There Welcome To our store again Thank You		
ST# AB123 TILL XXXX DR# 1 TRAN# 1010023 CSH: 1 07/19/13 18:06:56		

Receipt example for transaction with item level discount

RECEIPT WITH ITEM LEVEL DISCOUNT		
WELCOME TO OUR STORE VeriFone Gold Disk		
FL		
Description	Qty	Amount
T BREAD	5	49.85
LINE ITEM DISC		-8.75
T BUTTER	2	19.96
T MILK	6	59.94
LINE ITEM DISC		-15.00
Subtotal		106.00
Tax		26.50
TOTAL		132.50
CREDIT \$		132.50
MASTERCARD \$132.50 Acct/Card #: XXXXXXXXXXXXX1491 Auth #: 987654 Resp Code: 0 Stan: 0001116 Invoice #: 6 SITE ID: VP13007411001 MERCHANT COPY		
SIGNATURE I agree to pay the amount stated on this receipt.		
LINE 1 MESSAGE 1 Card Balance: Unknown Last line		
REWARD Acct/Card #: XXXXXXXXXXXXX6197 Trace #: 3 Stan: 0001116 Hi There Welcome To our store again Thank You		
ST# AB123 TILL XXXX DR# 1 TRAN# 1010021 CSH: 1 07/19/13 17:51:02		

3.16 EPS Loyalty Reports

This section details various loyalty related reports. Each of the following reports contains loyalty transaction details from current/existing LHFEF implementations. All the transactions in these reports will be sorted on the date and time of the transaction.

Loyalty PPU Transaction Details

This report contains information about the entire fuel sale with a loyalty PPU discount. This report would be sorted on transaction date and time and will be requested by Terminal Batch Number. For each sale this report will contain the date and time, fueling position, grade, original price as reported by POS, PPU discount amount, volume, total dollar amount of the discount, total amount of the fuel sale, and the loyalty card number.

Example:

LINE	TEXT
0	DESTROY AFTER USE DO NOT JUST DISCARD
1	Loyalty PPU Transaction Details
2	DLR # xxxxxxxx
3	Printed: xx/xx/xx xx:xx:xx:xx
4	
5	Terminal Batch XXX
6	Date Time Member-ID Pump# Grade Street-Price PPU-Disc Volume Total-Disc Total-Sale
7	ccyy-mm-dd hh:mm xxxxxxxxxxxxxxxxxxxxxx x xxx xxx.xxx xxx.xxx xxx.xxx xxx.xx xxx.xx
8	Total Volume x,xxx.xxx
9	Total Discount Amount x,xxx.xxx
10	Total Sale Amount x,xxx.xxx
11	
12	DESTROY AFTER USE DO NOT JUST DISCARD

Line-by-line description is given below.

0. Security warning
1. Title Line
2. Dealer Number
3. Contains the report printed information including both date and time
4. Skip line
5. Report terminal batch number
6. Column header
7. The transaction Details
 - Date – Date of the transaction (length 10 char)
 - Time – Time of the transaction (length 5 char)
 - Member ID – The customer loyalty ID (length Max 19)
 - Pump# - Pump number (length max 2)
 - Grade – The fuel grade (length max 3)
 - Street Price – The original price as reported by POS (max 7 char format xxx.xxx)
 - PPU Disc - The PPU Discount amount. (Max 7 chars format xxx.xxx)
 - Volume - The total volume of fuel dispensed. (Max 7 chars format xxx.xxx)
 - Total Disc - The total discount amount obtained for the transaction (max 6 char format xxx.xx)
 - Total Sale - Total amount of the fuel sale (max 6 char format xxx.xx)
8. Total Volume – Total Volume of Loyalty Fuel transactions during the period
9. Total Discount Amount – Totals for all Loyalty Discount transactions during the period
10. Total Sales Amount - Totals for all Loyalty Sale transaction amount during the period.
11. Skip Line
12. Security warning

Sample report is given below

Loyalty PPU Transaction Details Report

Date	Time	Member-ID	Pump#	Grade	Street-Price	PPU-Disc	Volume	Total-Disc	Total-Sale
2009-05-04	23:56	6032250003100000072	4	001	2.121	0.040	1.442	0.06	3.00
2009-05-04	23:56	6032250003100000072	15	003	2.121	0.040	1.442	0.06	3.00
Total Volume									2.884
Total Discount Amount									0.12
Total Sale Amount									6.00

Terminal Batch Loyalty Detail Report

The Loyalty Detail Report will be requested by Terminal Batch number. The report will contain the Loyalty Account Number, Total Amount of the transaction, and Discount Amount of the transaction. It will reference the Report Masking Flag in the EPS General Parameters Table to determine whether to use the masked account number. It will also include a sum of each of the total columns for that Terminal Batch.

It is critical that this report be able to be requested only with the proper security level because of the sensitive nature of the data. (Note: based on the parameter file settings, the account number may be masked.)

The report will be sorted on the Date and Time of the transaction.

Example:

LINE	TEXT		
0	D E S T R O Y A F T E R U S E D O N O T J U S T D I S C A R D		
1	Terminal Batch Loyalty Detail Report DLR # xxxxxxxx Printed: xx/xx/xx xx:xx:xx:xx Terminal Batch 999		
2	mm/dd/yy hh:mm:ss DLR# 99999999		
3	WORKSTATION ID xxx		
4			
4a	Loyalty Host:xxxx		
4b			
5	Account No.	Total	Discount
5a	99999999999999999999	999.99	999.99
5b	99999999999999999999	999.99-	999.99-
5c	5499999999999999	999.99	999.99
6	TICKET TOTAL		9999999.99
7	DISCOUNT TOTAL		9999999.99
4			
4a	Loyalty Host:xxxx		
4b			
5	Account No.	Total	Discount
5a	99999999999999999999	999.99	999.99
5b	99999999999999999999	999.99-	999.99-
5c	5499999999999999	999.99	999.99
6	TICKET TOTAL		9999999.99
7	DISCOUNT TOTAL		9999999.99
8			
9	Summary Discounts for all Loyalty Hosts		
10			

11	TICKET TOTAL	9999999.99
12	DISCOUNT TOTAL	9999999.99
13	D E S T R O Y A F T E R U S E D O N O T J U S T D I S C A R D	

Line-by-line description is given below.

0. Security warning
1. The title line should be centered in bold print
2. This line contains the date and time stamp, station number
3. Workstation ID

Lines 4-7 will be repeated for each Loyalty Host.

4. Skip Line
 - a) Loyalty Host Name
 - b) Skip Line
5. This section details each transaction in the batch with the card account number (which can be either masked or unmasked), card type, transaction reference number, and total sale amount. Note: this alignment shows the maximum size account number, 19 digits.
 - a) Example of a card length of 19 digits
 - b) If the transaction is an adjustment, append a minus sign after the amount
 - c) Example of a card length of 14 digits
6. Ticket Total: Calculated by summing all sales in the batch. (for that particular host)
7. Discount Total: Calculated by summing all discounts in the batch (for that particular host)
8. Skip Line
9. Section Title: Summary Discount for all Loyalty Hosts
10. Skip Line
11. Ticket Total: Calculated by summing all sales in the batch. (Inclusive of all hosts)
12. Discount Total: Calculated by summing all discounts in the batch. (Inclusive of all hosts)
13. Security warning.

Loyalty Discount by Type

This report will be requested by Terminal Batch number. The report contain a summary and grand totals for all PPG, transaction- level, and item-level discounts given during the selected terminal batch

This report will have the total of network loyalty discounts taken through the EPS.

Example:

Note	TEXT			
1	*** Loyalty Discounts By Type ***			
2	DLR # xxxxxxxx			
3	Printed: xx/xx/xx xx:xx:xx:xx			
4				
5	Terminal Batch 99			
6	PPU DISC	TICKET DISC	ITEM DISC	TOTAL \$
7	99999.99	99999.99	99999.99	99999999.99

Line-by-line description is given below.

1. Report Title
2. Dealer Number
3. Contains the report printed information including both date and time
4. Skip Line
5. Report Terminal Batch Number
6. Column Headers
7. Totals for all transactions during the period

Loyalty PPU Totals By Grade

This report will be requested by Terminal Batch number. For fuel sales with loyalty, the report will contain a list of fuel product codes sold, along with the total volume by product code and the total dollar amount of discounts by product code. The total number of discounted fuel transactions will also be reported. Fuel sales without loyalty will not be reported.

This report will have the total of network loyalty discounts taken by grade through the EPS.

Example:

Note	TEXT
1	*** Loyalty PPU Totals By Grade ***
2	DLR # xxxxxxxx
3	Printed: xx/xx/xx xx:xx:xx:xx
4	
5	Terminal Batch 99
6	GRADE COUNT VOLUME DISCOUNTS \$
7	999 9999 99999.99 99999.99
7	999 9999 99999.99 99999.99
7	999 9999 99999.99 99999.99

Line-by-line description is given below.

1. Report Title
2. Dealer Number
3. Contains the report printed information including both date and time
4. Skip Line
5. Report Terminal Batch Number
6. Column Headers
7. Totals for all transactions during the period

Line Item Discount Detail Report

This report will be requested by Terminal Batch number. Any sales with line item discount will be contained in this report. Each transaction will be an entry; there will be a column for transaction total and line item discount total for that particular entry. Transactions without line item discounts will not be reported.

Example:

LINE	TEXT
0	D E S T R O Y A F T E R U S E D O N O T J U S T D I S C A R D
1	Line Item Discount Detail Report DLR # xxxxxxxx Printed: xx/xx/xx xx:xx:xx:xx
2	Printed: mm/dd/yy hh:mm:ss
3	WORKSTATION ID xxx
4	
5	Terminal Batch 999
6	
6a	Loyalty Host:xxxx
6b	

7	Account No.	Total	Discount
7a	99999999999999999999	999.99	999.99
7b	99999999999999999999	999.99-	999.99-
7c	5499999999999999	999.99	999.99
8	NET TOTAL 9999999.99		
6			
6a	Loyalty Host:xxxx		
6b			
7	Account No.	Total	Discount
7a	99999999999999999999	999.99	999.99
7b	99999999999999999999	999.99-	999.99-
7c	5499999999999999	999.99	999.99
8	NET TOTAL 9999999.99 9999999.99		
9			
10	D E S T R O Y A F T E R U S E D O N O T J U S T D I S C A R D		

Line-by-line description is given below.

0. Security warning
1. The title line should be centered in bold print
2. This line contains the date and time stamp
3. Workstation ID
4. Skip Line
5. Terminal Batch Number for which the user requested the report.

Lines 6-9 will be repeated for each Loyalty Host.

6. Skip Line
 - a) Loyalty Host Name
 - b) Skip Line
7. This section details each transaction in the batch with the card account number (which can be either masked or unmasked), card type, transaction reference number, and total sale amount. Note: this alignment shows the maximum size account number, 19 digits.
 - a) Example of a card length of 19 digits
 - b) If the transaction is an adjustment, append a minus sign after the amount
 - c) Example of a card length of 14 digits
8. Net Total: Calculated by summing all transaction totals and line item discount totals in the batch. (for that particular host)
9. Skip Line
10. Security warning

3.17 Loyalty Card Velocity Checking

EPS Loyalty FEP Module will not implement the loyalty card velocity checking. The Loyalty host will be responsible for the loyalty card velocity checking.

3.18 Automated Exception Handling

EPS Loyalty FEP Module will reverse or cancel the transactions that have the loyalty discounts applied if instructed to do so by the EPS Core.

3.19 Diagnostics Page

The Diagnostics page will display FEP information for both payment and loyalty hosts. In case of multiple loyalty hosts the diagnostic page will display FEP information for all the Loyalty modules configured.

The POS section of the Diagnostics page will show whether each WorkstationID is loyalty capable.

3.20 PA-DSS Considerations

The EPS may send sensitive account information to the loyalty host; therefore the appropriate PCI standards should be reviewed and followed.

4. Configuration

4.1 Loyalty FEP Configuration

The configuration tool will have a loyalty configuration UI for each loyalty FEP (for example, “LFEP1 Configuration”) for the user to configure some of the parameters, such as Dealer ID, Program Name (custom name for a loyalty FEP), IP Address, Port, Loyalty Prompting, and Program Precedence on FEP Table.

4.2 Loyalty Card Configuration

The user can use the configuration tool to configure cards for an enabled/configured LFEP. For defining a new card, customers are expected to provide parameters such as Card Table Identifier (Unique ID), Card Name, ISP Range, Pan Length, and Manual Entry Allowed.

4.3 Loyalty Prompting Flag

This is a configurable parameter on an FEP Table that determines whether prompting for that particular LFEP will be done. For example, for a loyalty transaction on a multiple loyalty FEP enabled system, there will be prompts for the cards from different FEPs, such as “If you have cards of LFEP1/LFEP2/LFEP3/Payment, please swipe now.” If this flag is not enabled, that particular LFEP’s custom name does not appear in any card prompts.

4.4 Card Acquisition and Card Table Matching

The EPS can acquire customer card information from either the POS or the POP (PINPad, DCR keypad/reader, scanner, RFID reader, or mobile wallet) via a prompt. The loyalty ID/card must be configured in the Card Table.

Idle Prompt

With proper table configuration, at the idle prompt, the customer will be able to use an RFID or magnetic stripe loyalty card, payment card, or hybrid card, or indicate a desire to manually enter an alternate loyalty ID.

Hybrid Cards

A loyalty card identified as a hybrid card will be processed both as loyalty and payment provided no other cards have been previously presented. If a hybrid card is not the first card presented, it may be used as payment only.

Hybrid cards have two entries in the Card Table. A hybrid card's loyalty entry should always precede its payment entry. Processing of the hybrid card by each FEP is independent of the other FEP.